



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

NEET(UG)

MINOR

04-03-2023

ENTHUSIAST ADVANCE COURSE PHASE : MEF-I (B)

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

This Booklet contains 28 pages.

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on the Back Cover of this Test Booklet.

Important Instructions :

- On the Answer Sheet, fill in the particulars on **Side-1** and **Side-2** carefully with **blue/black** ball point pen only.
- The test is of **3 Hours 20 Minutes** and this Test Booklet contains **200** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are **720**.
- In this Test Paper, each subject will consist of **two sections**. **Section A** will consist of **35** questions (all questions are mandatory) and **Section B** will have **15** questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.
- In case of more than one option correct in any question, the best correct option will be considered as answer.
- Use **Blue/Black Ball Point Pen only** for writing particulars on this page/markings responses.
- Rough work is to be done on the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.**
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is **not** permissible on the Answer Sheet.

Name of the Candidate (in Capitals) _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ Invigilator's Signature : _____

Your Target is to secure Good Rank in Pre-Medical 2024

Topic : GRAVITATION, NLM, FRICTION, ELECTROSTATICS (TILL COVERED IN CLASS)

SECTION - A

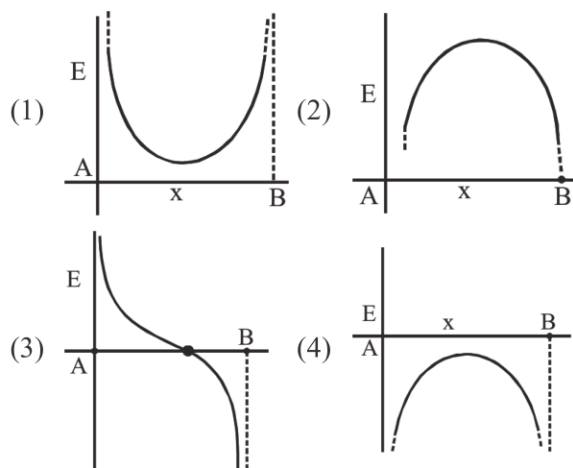
TG: @ChalnaayaaarYour text here

Attempt All 35 questions

1. body has -80 microcoulomb of charge. Number of additional electrons on it will be :

- (1) 8×10^{-5}
 (2) 80×10^{15}
 (3) 5×10^{14}
 (4) 1.28×10^{-17}

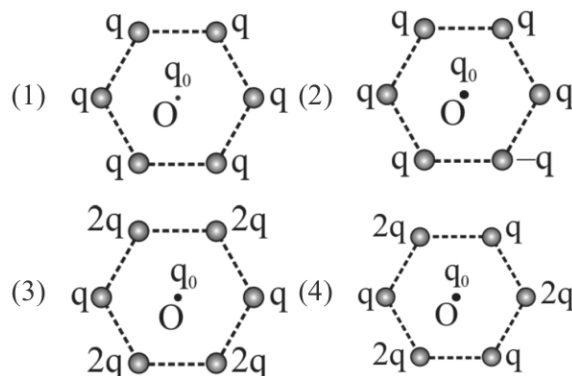
2. Two positive point charges q_1 & q_2 ($|q_1| > |q_2|$) are placed at point A & B at 'x' separation draw graph showing variation of electric field intensity with distance from 'A' :-



3. If $q_1 + q_2 = q$, then the value of the ratio $\frac{q_1}{q}$, for which the force between q_1 and q_2 is maximum is:

- (1) 0.25
 (2) 0.75
 (3) 1
 (4) 0.5

Figures below show regular hexagons, with charges at the vertices. In which of the following cases the net electric force at the centre is not zero :-



5. Twelve point charges each of charge q coulomb are placed at the circumference of a circle of radius r with equal angular spacing. If one of the charges is removed, the net electric field (in N/C) at the centre of the circle is : (ϵ_0 – permittivity of free space)

- (1) $\frac{13q}{4\pi\epsilon_0 r^2}$
 (2) zero
 (3) $\frac{q}{4\pi\epsilon_0 r^2}$
 (4) $\frac{12q}{4\pi\epsilon_0 r^2}$

6. Two point charges repel each other with a force of 100 N. One of the charges is increased by 10% and the other is reduced by 10% . The new force of repulsion at the same separation would be :

- (1) 100 N (2) 121 N
 (3) 99 N (4) 200 N

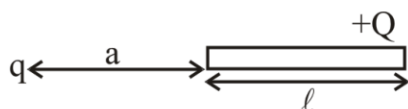
7. **Statement-1** : Two positively charged bodies can attract each other.

Statement-2 : Flux through a gaussian surface depends on its shape.

- (1) Statement-1 is correct, statement-2 is incorrect.
 (2) Statement-1 is incorrect, statement-2 is correct.
 (3) Both statements are correct.
 (4) Both statements are incorrect.
8. Volume charge density of a solid sphere of radius R is given by $\rho = \rho_0 r$ here r is distance from centre. Find charge present on sphere.

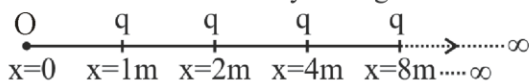
- (1) $\frac{\rho_0 \pi R^4}{4}$ (2) $\rho_0 \pi R^3$
 (3) $\rho_0 \pi R^4$ (4) $\frac{\rho_0 \pi R^3}{3}$

9. Find the electrostatic force between a point charge and rod as shown :



Rod is uniformly charged.

- (1) $\frac{kQ}{a(a-l)}$ (2) $\frac{kQ}{a(a+l)}$
 (3) $\frac{kqQ}{a(a-l)}$ (4) $\frac{kqQ}{a(a+l)}$
10. Find electric field intensity at origin.



- (1) $\frac{4}{5} kq$ (2) $\frac{4}{3} kq$ (3) $\frac{kq}{4}$ (4) $\frac{4kq}{2}$
11. A cube of side 10 cm is placed in an electric field of $5 \times 10^3 \hat{i} \text{ NC}^{-1}$. Find net flux passing through the cube.
- (1) Zero (2) $50 \text{ Nm}^2 \text{C}^{-1}$
 (3) $100 \text{ Nm}^2 \text{C}^{-1}$ (4) $25 \text{ Nm}^2 \text{C}^{-1}$

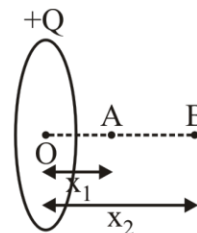
12. Consider a cube of side a . $2C$ charge is placed at its centre and $1C$ charges are placed at mid point of each edge. Find net flux of the cube.

- (1) $\frac{5}{\epsilon_0}$ (2) $\frac{10}{\epsilon_0}$ (3) $\frac{20}{\epsilon_0}$ (4) $\frac{25}{\epsilon_0}$

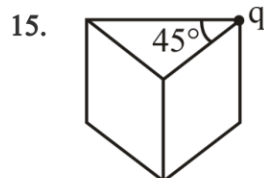
13. Two equal point charges A and B are R distance apart. A third point charge placed on the perpendicular bisector at a distance x from the centre will experience a maximum electrostatic force when :-

- (1) $x = \frac{R}{2\sqrt{2}}$ (2) $x = \frac{R}{\sqrt{2}}$
 (3) $x = R\sqrt{2}$ (4) $2\sqrt{2}R$

14. Electric field due to a uniformly charged ring on axis at a point A very close to centre is same as at point B far away from centre. The radius of ring is :



- (1) $\sqrt{x_1 x_2}$ (2) $(x_1^2 x_2)^{1/3}$
 (3) $(x_2^2 x_1)^{1/3}$ (4) $x_1 + x_2$



15. A prism shaped imaginary structure is shown. A point charge is kept as shown. Calculate electric flux passing through the prism.

- (1) $\frac{q}{32\epsilon_0}$ (2) $\frac{q}{16\epsilon_0}$
 (3) $\frac{q}{8\epsilon_0}$ (4) $\frac{q}{4\epsilon_0}$

16. The ratio of coulomb's electrostatic force to the gravitational force between an electron and a proton separated by some distance is 2.4×10^{39} . The ratio of the proportionality constant, $K = \frac{1}{4\pi\epsilon_0}$ to the Gravitational constant G is nearly (Given that the charge of the proton and electron each = 1.6×10^{-19} C, the mass of the electron = 9.11×10^{-31} kg, the mass of the proton = 1.67×10^{-27} kg):

(1) 10^{20} (2) 10^{30} (3) 10^{40} (4) 10

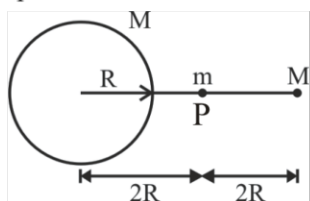
17. Two metallic spheres of radii R_1 and R_2 are connected by a thin wire. If $+q_1$ and $+q_2$ are the charges on the two spheres, then :

(1) $\frac{q_1}{q_2} = \frac{R_1^2}{R_2^2}$ (2) $\frac{q_1}{q_2} = \frac{R_1}{R_2}$
 (3) $\frac{q_1}{q_2} = \frac{R_1^3}{R_2^3}$ (4) $\frac{q_1}{q_2} = \frac{(R_1^2 - R_2^2)}{(R_1^2 + R_2^2)}$

18. Suppose the gravitational force varies inversely as the n^{th} power of the distance. Then the time period of a planet in circular orbit of radius R around the sun will be proportional to :

(1) R^n (2) $R^{(n+1)/2}$
 (3) $R^{(n-1)/2}$ (4) R^{-n}

19. Find the gravitational potential energy of mass m placed at point P as shown :



(1) $\frac{-GMm}{R}$ (2) $\frac{-GMm}{2R}$
 (3) $\frac{-GMm}{4R}$ (4) $\frac{-2GMm}{R}$

20. The depth from the surface of the earth at which the effective value of acceleration due to gravity is $\frac{g}{4}$ is (R_e = radius of earth)

(1) R_e (2) $\frac{3R_e}{4}$ (3) $\frac{R_e}{2}$ (4) $\frac{R_e}{4}$

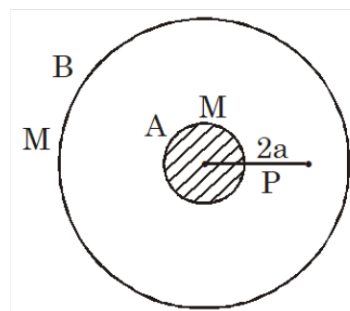
21. Two point masses having mass M and $4M$ are placed at distance r . The gravitational potential at a point, where gravitational field intensity is zero, is :-

(1) $-\frac{9GM}{r}$ (2) $-\frac{2GM}{3r}$
 (3) $-\frac{3GM}{r}$ (4) $-\frac{6GM}{5r}$

22. What should be angular speed with which the earth have to rotate on its axis so that a person on equator would weigh $\frac{3}{5}$ th as much as his present actual weigh (If acceleration due to gravity at surface is g & radius of earth is R) :-

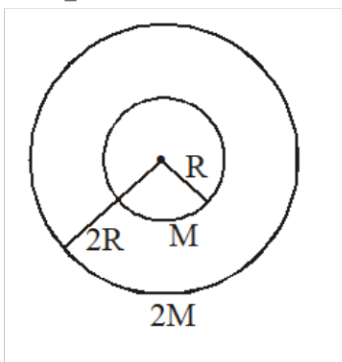
(1) $\sqrt{\frac{2g}{5R}}$ (2) $\sqrt{\frac{2R}{5g}}$ (3) $2\sqrt{\frac{R}{5g}}$ (4) $\frac{2g}{5R}$

23. In the given situation 'A' is a solid sphere of mass M and radius a . 'B' is a spherical shell of same mass M with radius $3a$. The gravitational Intensity at a point 'P' which is situated at a distance $2a$ from their's common centre.



(1) $\frac{2GM}{4a^2}$ (2) $\frac{GM}{4a^2}$ (3) $\frac{GM}{9a^2}$ (4) 0

24. Two concentric hollow spheres of mass $2M$ and M having radius $2R$ and R respectively are placed as shown then gravitational potential at a distance of $\frac{3R}{2}$ from centre is :-



- (1) $-\frac{GM}{R}$ (2) $-\frac{2GM}{3R}$
 (3) $-\frac{5GM}{3R}$ (4) $-\frac{GM}{3R}$
25. A particle is thrown with a speed $v = \sqrt{\frac{gR}{2}}$ from the surface of the earth then find the speed of particle at height $h = \frac{R}{4}$ from earth's surface ?

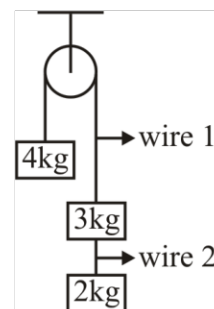
- (1) $\sqrt{\frac{gR}{10}}$ (2) $\sqrt{\frac{gR}{20}}$
 (3) $\sqrt{\frac{11}{10}gR}$ (4) None of these

26. When 10 wagons of trains are pulled by force of 10^5 ; if mass of each wagon is 1500 kg then find force exerted on last four wagon ?

- (1) 2×10^4 N
 (2) 4×10^4 N
 (3) 3×10^4 N
 (4) 5×10^4 N

TG: @Chalnaayaaar

27. Calculate tension in wire 1 shown in figure :-

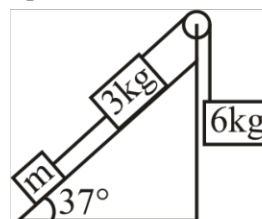


- (1) $\frac{400}{9}$ N (2) $\frac{200}{9}$ N
 (3) $\frac{250}{9}$ N (4) $\frac{500}{9}$ N

28. The ratio of weights of a man in a stationary lift and in a lift accelerating downward with a uniform acceleration is 5 : 4. The acceleration of the lift is :-

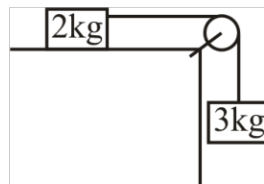
- (1) $\frac{g}{3}$ (2) $\frac{g}{2}$ (3) g (4) $\frac{g}{5}$

29. If system is in equilibrium then find value of mass m ?



- (1) 10 kg (2) 3 kg
 (3) 7 kg (4) 6 kg

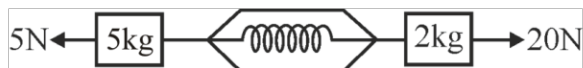
- 30.



In the given figure find out distance travelled by 2 kg block in first 2 sec ?

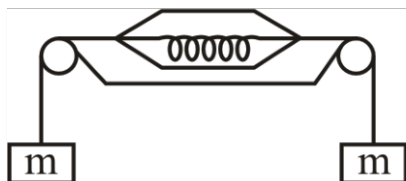
- (1) 10 m (2) 12 m
 (3) 24 m (4) 8 m

31. In given figure calculate reading of spring balance ?



- (1) $\frac{110}{7}$ N (2) $\frac{50}{7}$ N
(3) $\frac{150}{7}$ (4) 150 N

32. Calculate reading of spring balance when system is in equilibrium ?



- (1) m (2) 2m
(3) m/2 (4) zero

33. Calculate tension in string ?



- (1) 18 N (2) 6 N
(3) 12 N (4) 10 N

34. Find acceleration of 3 kg in given figure ?



- (1) 2 m/s^2 (2) 4 m/s^2
(3) 12 m/s^2 (4) 10 m/s^2

35. If force $4\hat{i} + 4\hat{j} + 7\hat{k}$ acts on body and produces acceleration 3m/s^2 then find mass of body ?

- (1) 4 kg
(2) 3 kg
(3) 5 kg
(4) 7 kg

SECTION - B

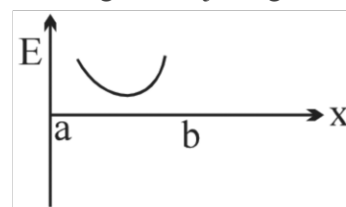
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This section will have 15 questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

36. The dimension of ϵ_0 is :

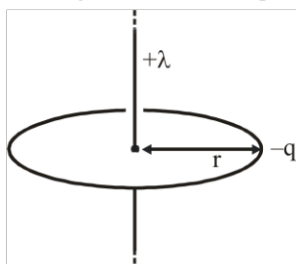
- (1) $[M^{-1} L^{-3} T^4 A^2]$
(2) $[M^0 L^{-3} T^2 A^3]$
(3) $[M^{-1} L^{-3} T^3 A]$
(4) $[M^{-1} L^{-3} T A^2]$

37. Two point charges a & b whose magnitudes are same are positioned at a certain distance from each other; a is at origin. Graph is drawn between electric field strength E and distance x from a. E is taken positive if it is along the line joining from a to b



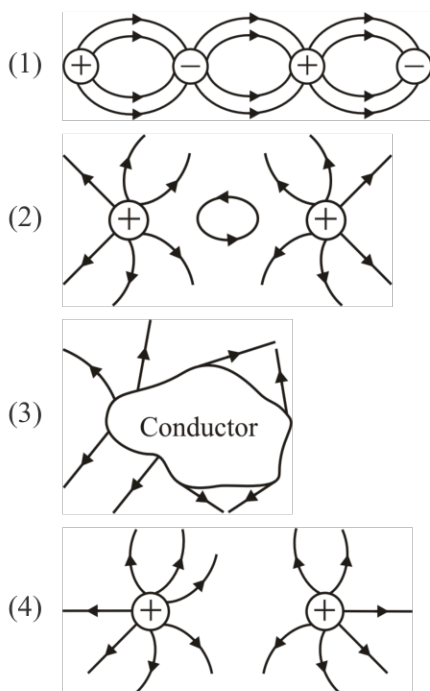
- (1) a is positive, b is negative
(2) a & b both are positive
(3) a & b both are negative
(4) a is negative, b is positive

38. A particle of charge $-q$ & mass m moves in a circle of radius r around an infinitely long line charge on linear charge density $+\lambda$. Then time period will be

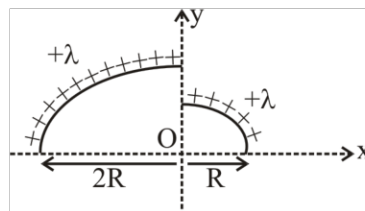


- (1) $T = 2\pi r \sqrt{\frac{m}{2k\lambda q}}$
 (2) $T^2 = \frac{4\pi^2 m}{2k\lambda q} r^3$
 (3) $T = \frac{1}{2\pi r} \sqrt{\frac{2k\lambda q}{m}}$
 (4) $T = \frac{1}{2\pi r} \sqrt{\frac{m}{2k\lambda q}}$

39. Which of the following electrostatic field lines are possible ?



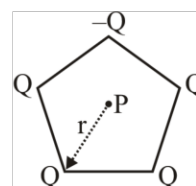
- 40.



Find electric field intensity at origin due to two circular arcs shown.

- (1) $\frac{2k\lambda}{R}$ (2) $\frac{\sqrt{10}}{2} \frac{k\lambda}{R}$
 (3) $\frac{k\lambda}{R}$ (4) $\frac{\sqrt{5}}{4} \frac{k\lambda}{R}$

41. Find electric field intensity at centre of the regular pentagon.



- (1) $\frac{kQ}{r^2}$
 (2) $\frac{2kQ}{r^2}$
 (3) $\frac{5kQ}{r^2}$
 (4) $\frac{6kQ}{r^2}$

42. Two identical charged spheres suspended from a common point by two massless strings of length ℓ are mutually at a distance d ($d \ll \ell$) apart because of their mutual repulsion. The charges begin to leak from both the spheres at a constant rate. As a result, the spheres approach each other with a velocity v . The v varies as a function of distance x between the spheres, as :

- (1) $v \propto x^{-1}$ (2) $v \propto x^{1/2}$
 (3) $v \propto x$ (4) $v \propto x^{-1/2}$

43. Two charged spherical conductors of radius R_1 and R_2 are connected by a wire. Then the ratio of surface charge densities of the spheres (σ_1/σ_2) is :

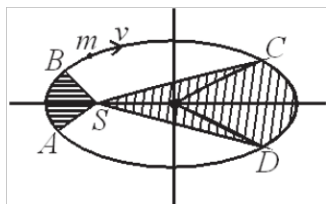
(1) $\frac{R_1}{R_2}$ (2) $\frac{R_2}{R_1}$
 (3) $\sqrt{\left(\frac{R_1}{R_2}\right)}$ (4) $\frac{R_1^2}{R_2^2}$

TG: @Chalnaayaaar

44. A satellite moves around the earth in a circular orbit of radius r with speed v . If the mass of the satellite is M , its total energy is:

(1) $-\frac{1}{2}Mv^2$ (2) $\frac{1}{2}Mv^2$
 (3) $\frac{3}{2}Mv^2$ (4) Mv^2

45. The figure shows elliptical orbit of a planet m about the sun S . The shaded area SCD is twice the shaded area SAB . If t_1 is the time for the planet to move from C to D and t_2 is the time to move from A to B , then



(1) $t_1 < t_2$ (2) $t_1 = 4t_2$
 (3) $t_1 = 2t_2$ (4) $t_1 = t_2$

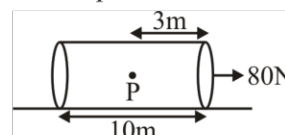
46. A projectile is projected with velocity $k v_e$ in vertically upward direction from ground into space (v_e is escape velocity and $k < 1$). If air resistance is considered negligible then find the maximum height from centre of earth to which it can go.

(1) $\frac{R}{k^2 + 1}$ (2) $\frac{R}{k^2 - 1}$
 (3) $\frac{R}{1 - k^2}$ (4) $\frac{R}{k + 1}$

47. A particle is projected vertically upwards from the surface of the earth (radius R_e) with a speed equal to one fourth of escape velocity. What is the maximum height attained by it from the surface of the earth ?

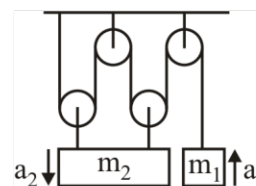
(1) $\frac{16}{15}R_e$ (2) $\frac{R_e}{15}$
 (3) $\frac{4}{15}R_e$ (4) None

48. Find out tension at point P :-



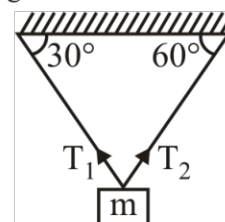
(1) 56 N (2) 24 N
 (3) 40 N (4) $\left(\frac{80}{3}\right)$ N

49. In the given figure calculate relation between acceleration of block A & B ?



(1) $a_1 = 4a_2$ (2) $4a_1 = a_2$
 (3) $2a_1 = a_2$ (4) $a_1 = 2a_2$

50. If system is in equilibrium then find out T_1 & T_2 if $m = 10$ kg



(1) 50 N, $50\sqrt{3}$ (2) 60 N, $60\sqrt{3}$
 (3) 50 N, 60 N (4) $50\sqrt{3}$, 60

Topic : CHEMICAL KINETIC, LIQUID SOLUTIONS (TILL TAUGHT), IUPAC, ISOMERISM (UPTO OPTICAL)

SECTION-A

Attempt All 35 questions

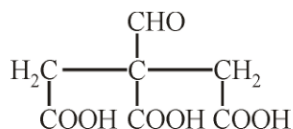
51. How many number of carbon present in second member of alkenyne :

- (1) 5 (2) 6 (3) 7 (4) 4

52. General formula of Alkanamine is :

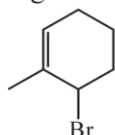
- (1) $C_nH_{2n+1}N$ (2) $C_nH_{2n+3}N$
(3) $C_nH_{2n-3}N$ (4) $C_nH_{2n-1}N$

53. IUPAC Name ?



- (1) 2-oxo propane-1,2,3-trioic acid
(2) 2-formyl propane-1,2,3-tricarboxylic acid
(3) 2-formyl propane-1,2,3-trioic acid
(4) 2-oxo propane-1,2,3-tricarboxylic acid

54. The IUPAC name of given compound is :

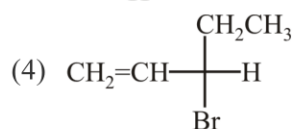
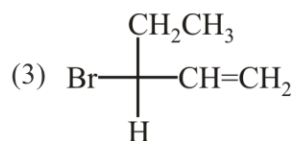
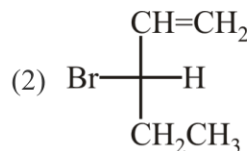
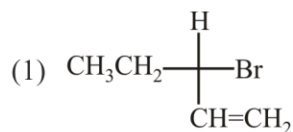


- (1) 2-Bromo-1-methyl cyclohexene
(2) 6-Bromo-1-methyl cyclohexene
(3) 1-Bromo-2-methyl cyclohexene
(4) 1-Bromo-2-methyl cyclohex-2-ene

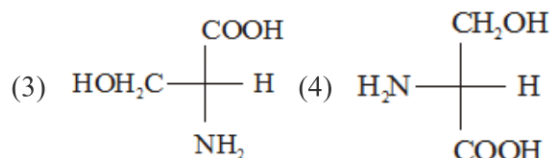
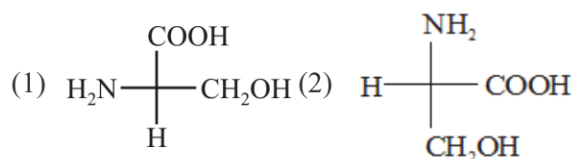
55. Which of the following biphenyl is optically active ?

- (1) (2)
(3) (4) None

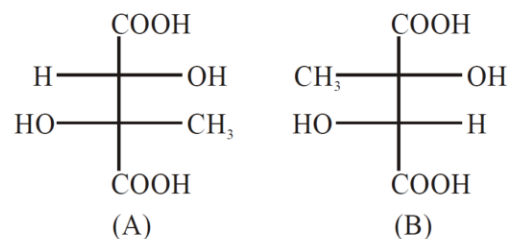
56. Which of the following compounds has an R configuration ?



57. Among the following L-serine is



58. The structures A and B represent :-

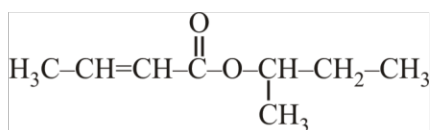


- (1) enantiomers (2) diastereomers
(3) homomers (4) racemic mixture

59. Which of the following is optically active :-

- (1)
- (2)
- (3)
- (4)

60. Common name ?

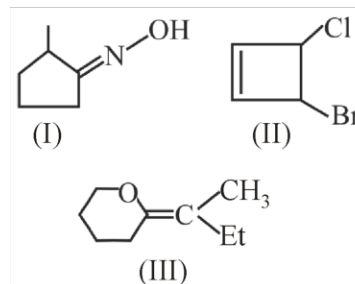


- (1) Isobutyl crotonate (2) Isobutyl butyrate
(3) Tert butyl acrylate (4) Sec-butyl crotonate

61. Which of the following molecule is achiral ?

- (1)
- (2)
- (3)
- (4)

62. How many compounds show GI ?



- (1) 1 (2) 2
(3) 3 (4) All of the above

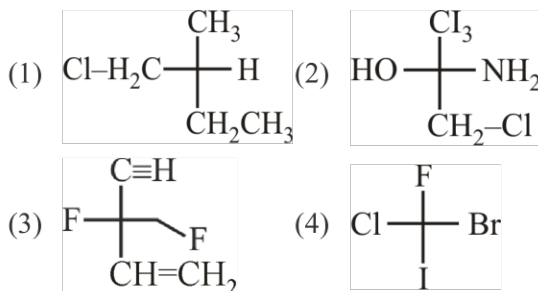
63. Which of the following statement is correct ?

- (1) Acetone and acetaldehyde are homologues
(2) and are functional group isomers.
(3) and are position isomers
(4) Trimethyl amine ethylmethyl amine are functional group isomers.

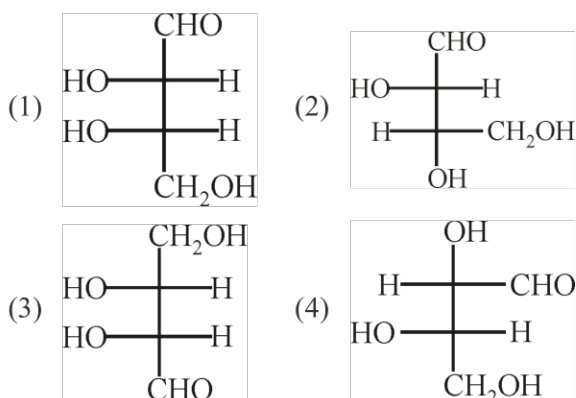
64. Which of the following is E isomer ?

- (1)
- (2)
- (3)
- (4)

65. Which of the following represents 'R' ?



66. Which of the following represent 'D' form ?



67. Total structural isomers of $\text{C}_4\text{H}_{11}\text{N}$?

- (1) 5 (2) 6
(3) 7 (4) 8

68. The rate of a reaction is expressed as:

$$+\frac{1}{2} \frac{\Delta[\text{C}]}{\Delta t} = \frac{1}{3} \frac{\Delta[\text{D}]}{\Delta t} = \frac{1}{4} \left(-\frac{\Delta[\text{A}]}{\Delta t} \right) = \left(-\frac{\Delta[\text{B}]}{\Delta t} \right)$$

The reaction is

- (1) $4\text{A} + \text{B} \rightarrow 2\text{C} + 3\text{D}$ (2) $\text{B} + 3\text{D} \rightarrow 4\text{A} + 2\text{C}$
(3) $\text{A} + \text{B} \rightarrow \text{C} + \text{D}$ (4) $\text{B} + \text{D} \rightarrow \text{A} + \text{C}$

69. The rate constant for a first order decomposition reaction is given by

$$\log K = 10 - \frac{1000}{T}. \text{ Then, what will be activation energy in kcal/mol ?}$$

- (1) 4.60 (2) 17.6
(3) 3.2 (4) 9.8

70. Molecularity of a reaction -

- (1) is always positive integer number
(2) cannot be less than 2
(3) can have a fractional value
(4) can be zero

71. Observe the following reaction, $2\text{A} + \text{B} \rightarrow \text{C}$

The rate of formation of C is $2.2 \times 10^{-3} \text{ mol L}^{-1} \text{ min}^{-1}$.

What is the value of $-\frac{d[\text{A}]}{dt}$ (in $\text{mol L}^{-1} \text{ min}^{-1}$)?

- (1) 2.2×10^{-3} (2) 1.1×10^{-3}
(3) 4.4×10^{-3} (4) 5.5×10^{-3}

72. Which of the following rate law has an overall order of 0.5 for reaction involving substances x, y and z?

- (1) $\text{Rate} = K (\text{C}_x) (\text{C}_y) (\text{C}_z)$
(2) $\text{Rate} = K (\text{C}_x)^{0.5} (\text{C}_y)^{0.5} (\text{C}_z)^{0.5}$
(3) $\text{Rate} = K (\text{C}_x)^{1.5} (\text{C}_y)^{-1} (\text{C}_z)^0$
(4) $\text{Rate} = K (\text{C}_x) (\text{C}_z)^0 / (\text{C}_y)^2$

73. For a reaction $\text{A} + \text{B} \rightarrow \text{P}$ the rate of the reaction at various concentrations are given below and it's rate law is given as $r = K[\text{A}]^a [\text{B}]^b$, value of a and b will be:-

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Experiment No.	[A]	[B]	Rate ($\text{mol dm}^{-3} \text{ s}^{-1}$)
1	0.2	0.2	0.04
2	0.1	0.4	0.04
3	0.2	0.4	0.08

- (1) $a = 1, b = 1$ (2) $a = 2, b = 0$
(3) $a = 2, b = 1$ (4) None

74. For a reaction $A \xrightarrow{k_r=0.6\text{M min}^{-1}} 2B$ starting with 1 M of 'A' only, concentration of B (in M) after 100 sec. is ?
 (1) 2 M (2) 1 M (3) 3 M (4) 0 M
75. A 1st order reaction is 75% completed in 100 minutes. How long time will it take for 50% completion?
 (1) 150 min
 (2) 200 min
 (3) 50 min
 (4) 75 min
76. For the reaction $\text{H}_2(\text{g}) + \text{Br}_2(\text{g}) \rightarrow 2\text{HBr}(\text{g})$, the experimental data suggests, $\text{Rate} = K[\text{H}_2][\text{Br}_2]^{1/2}$. The molecularity and order of reaction for the reaction is –
 (1) 2 and 2 respectively
 (2) 2 and 1½ respectively
 (3) 1½ and 2 respectively
 (4) 1½ and 1½ respectively
77. When temperature of $A \rightarrow P$ reaction increases from 300 to 310 K then its rate of reaction becomes two times, the activation energy of this reaction will be–
 (1) 53.6 kJ mol⁻¹ (2) 42.88 kJ mol⁻¹
 (3) 35.88 kJ mol⁻¹ (4) 60 kJ mol⁻¹
78. Which of the following represent temperature coefficient (μ) :-
 (1) $\frac{K_{T=45^\circ}}{K_{T=35^\circ}}$ (2) $\frac{K_{T=35^\circ}}{K_{T=20^\circ}}$
 (3) $\frac{K_{T=50^\circ}}{K_{T=30^\circ}}$ (4) $\frac{K_T}{K_{T+10}}$
79. Which of the following is correct expression of Arrhenius equation-
 (1) $r = ke^{-\frac{E_a}{RT}}$
 (2) $\log(k) = \log(A) - \frac{E_a}{RT}$
 (3) $\ln(k) = \ln(A) - \frac{E_a}{RT}$
 (4) $\ln(k) = \ln(A) - \frac{E_a}{2.303RT}$
80. Which of the following factor can alter activation energy (E_a) ?
 (1) Temperature (2) Reaction stoichiometry
 (3) Catalyst (4) Both (1) and (3)
81. For 1st order reaction unit of Arrhenius factor (A) will be
 (1) Ms⁻¹ (2) S⁻¹
 (3) M⁻¹s (4) M⁻¹ s⁻¹
82. Which of the following concentration term is temperature dependent ?
 (1) Mole fraction (2) Molarity
 (3) Molality (4) Mass percentage
83. What will be the normality of 2M H₃PO₂ (aq) solution ?
 (1) 2N (2) 4N (3) 1N (4) 6N
84. What will be the molarity (M) of NaCl in final solution if equal volumes of 0.01 M of HCl(aq) solution and 0.01M of NaOH(aq) solutions are mixed together ?
 (1) 0.01 M (2) 0.05 M
 (3) 0.005 M (4) 0.02 M

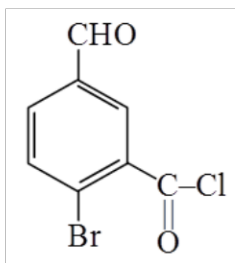
85. In the presence of catalyst which of the following parameter remains unchanged

- (1) E_a (activation energy)
- (2) ΔH
- (3) Rate of reaction
- (4) value of rate Constant (k)

SECTION-B

This section will have 15 questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

86. The IUPAC name of the following compound is :-



- (1) 4-bromo-3-chlorobenzaldehyde
- (2) 2-bromo-5-formyl benzene carbonylchloride
- (3) 2-chloro carbonyl bromobenzene
- (4) 6-bromo-3-formylbenzo-3-yl chloride

87. Which of the following pair are not Homologues ?

- (1) $\text{CH}_3\text{--CH}_2\text{--CH}_2\text{--NH}_2$ and $\text{CH}_3\text{--CH(NH}_2\text{)--CH}_3$
- (2) $\text{CH}_3\text{--C(=O)OH}$ and $\text{CH}_3\text{--C(=O)CH}_2\text{OH}$
- (3) $\text{CH}_3\text{--CH}_2\text{OH}$ and $\text{CH}_3\text{--CH}_2\text{OCH}_3$
- (4) All of these

88. Which of the following IUPAC name is incorrect ?

- (1) 3,6-Dimethylcyclohexene
- (2) 1, 6-Dimethylcyclohexene
- (3) 6,6-Dimethylcyclohexene
- (4) 1,5-Dimethylcyclohexene

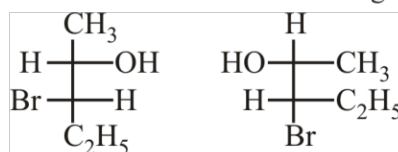
89. Which of the following compound can show geometrical isomerism?

- (1)
- (2)
- (3)
- (4)

90. Which structure are optically inactive due to only center of symmetry or COS

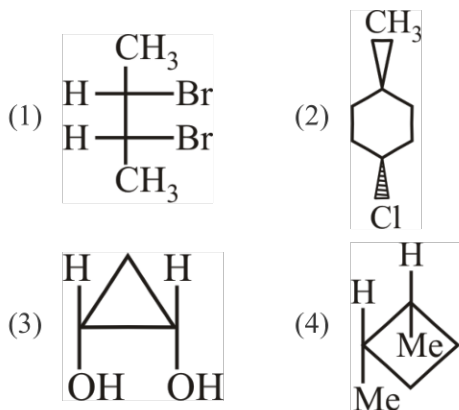
- (1)
- (2)
- (3)
- (4)

91. Relation between the following compounds.



- (1) Identical
- (2) Enantiomers
- (3) Diastereomers
- (4) Epimers

92. Which of the following is not a meso compound ?



93. Select incorrect statement :

- (1) All geometrical isomers are diastereomers
- (2) Meso compound is optically inactive due to internal compensation.
- (3) If a molecule has zero chiral center, then it will always be optically inactive
- (4) Racemic mixture is optically inactive due to external compensation.

94. The rate constant, the activation energy and the frequency factor of a chemical reaction at 25°C are $3.0 \times 10^{-4} \text{ s}^{-1}$, $104.4 \text{ KJ mol}^{-1}$ and $6.0 \times 10^{14} \text{ s}^{-1}$ respectively. The value of the rate constant as $T \rightarrow \infty$ is :

- (1) $2.0 \times 10^{18} \text{ s}^{-1}$
- (2) $6.0 \times 10^{14} \text{ s}^{-1}$
- (3) ∞
- (4) $3.6 \times 10^{30} \text{ s}^{-1}$

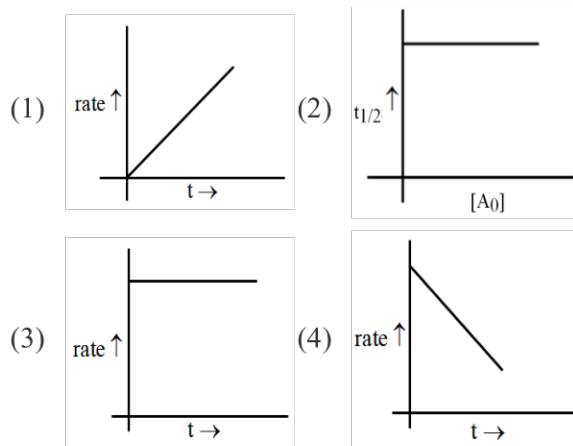
95. For which of the following, the units of rate constant and rate of the reaction are same -

- (1) First order reaction
- (2) Second order reaction
- (3) Third order reaction
- (4) Zero order reaction

96. The rate constant of a first order reaction is 3×10^{-6} per second. If the initial concentration is 0.10 M , the initial rate of reaction is _____.

- (1) $3 \times 10^{-5} \text{ Ms}^{-1}$
- (2) $3 \times 10^{-6} \text{ Ms}^{-1}$
- (3) $3 \times 10^{-8} \text{ Ms}^{-1}$
- (4) $3 \times 10^{-7} \text{ Ms}^{-1}$

97. Which of the following graph is correct for zero order reaction ?



98. What will be the molarity (M) of 2m acetic acid (CH_3COOH) solution ($d_{\text{sol}} = 1.12 \text{ g/ml}$) ?

- (1) 2M
- (2) 1.9M
- (3) 2.1M
- (4) 3M

99. If $t_{1/2}$ for a first order reaction is 10 min its $t_{90\%}$ will be

- (1) 30 min
- (2) 40 min
- (3) 33.3 min
- (4) 20 min

100. Which of the following relation is correct for 1^{st} order reaction

- (1) $t_{1/2} = \frac{\log(2)}{k}$
- (2) $t_{100\%} = 2 \times t_{50\%}$
- (3) $t_{87.5\%} = 3 \times t_{75\%}$
- (4) $\frac{t_{87.5\%}}{t_{75\%}} = \frac{3}{2}$

SUBJECT : BIOLOGY-I

Topic : HUMAN HEALTH AND DISEASE (COMPLETE), EVOLUTION (TILL TAUGHT), HUMAN REPRODUCTION, ANIMAL HUSBANDRY

SECTION-A

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Attempt All 35 questions

101. Best breed of chicken in layer category is :-

- (1) Plymouth rock (2) Aseel
(3) Leghorn (4) Rhode island red

102. Which breeding method is used to overcome inbreeding depression :-

- (1) Out crossing (2) Inbreeding
(3) Artificial insemination (4) All of the above

103. Mark the correct :-

(1)	Pashmina	(a)	Sheep
(2)	Stallion	(b)	Sterile male
(3)	Holstein-friesian	(c)	Exotic breed of cow
(4)	Angoor wool	(d)	Chiru

104. A patient is advised to intake omega-3 fatty acid in his/her diet, he/she should eat which of the following fishes :-

- (1) Pomfret (2) Catla
(3) Rohu (4) Mrigal

105. **Assertion** : Outcrossing is the best breeding method for animals that are below average in productivity in milk production, growth rate in beef cattle etc.

Reason : Outcrossing increases the heterozygosity.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

106. In MOET technology:-

- (1) Genetic mother produce one egg per cycle.
(2) Genetic mother is administered with FSH like hormone.
(3) Fertilized eggs are recovered at 64 cells stage to transfer into surrogate mother
(4) New offsprings have the characters of surrogate mother.

107. **Statement-I** : Hisardale is a new breed of sheep, developed by cross breeding method.

Statement-II : Cross breeding is done between animals of different breeds.

- (1) Statement-I correct but statement-II incorrect.
(2) Statement-II correct but statement-I incorrect.
(3) Both statement-I and II are correct.
(4) Both statement-I and II are incorrect.

108. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) :

Breast-feeding during the initial period of infant is recommended by doctors.

Reason (R): The milk produced during the initial few days of lactation contains several antibodies absolutely essential to develop resistance for the new-born babies.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
(3) (A) is correct but (R) is not correct
(4) (A) is not correct but (R) is correct

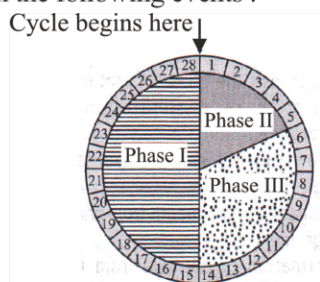
109. In the absence of fertilization, corpus luteum:

- (1) Stops secreting progesterone but increases the secretion of estrogen
- (2) Stops secreting estrogen but increases the secretion of progesterone
- (3) Increases the secretion of both progesterone and estrogen
- (4) Degenerates

110. A woman could still have menstrual periods after the surgical removal of:-

- (1) Both ovaries
- (2) One ovary only
- (3) One ovary and the uterus
- (4) Uterus only

111. Refer the given schematic representation of menstrual cycle showing three phases I, II and III. Select the option that correctly matches these phases with the following events :-



- (i) FSH secreted by the anterior pituitary stimulates the ovarian follicle to secrete oestrogen which stimulates the proliferation of endometrium.
- (ii) LH secreted by the anterior pituitary stimulates the development of corpus luteum which secretes progesterone
- (iii) Reduced production of LH from anterior pituitary causes degeneration of corpus luteum, thereby menstrual flow starts.

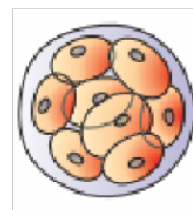
	Phase I	Phase II	Phase III
(1)	(i)	(ii)	(iii)
(2)	(iii)	(i)	(ii)
(3)	(ii)	(iii)	(i)
(4)	(iii)	(ii)	(i)

112. Match the column-A and B about the embryonic development of human :-

	Column-A		Column-B
(i)	End of one month	(a)	Most of the major organ systems developed
(ii)	End of second month	(b)	Appearance of hair on head
(iii)	End of three month	(c)	Heart formation
(iv)	During fifth month	(d)	Eye lids separated
(v)	End of sixth month	(e)	Limbs and digits formation

- (1) (i)-e, (ii)-b, (iii)-c, (iv)-a, (v)-d
- (2) (i)-e, (ii)-c, (iii)-b, (iv)-b, (v)-d
- (3) (i)-c, (ii)-e, (iii)-b, (iv)-a, (v)-d
- (4) (i)-c, (ii)-e, (iii)-a, (iv)-b, (v)-d

113. Observe the given diagram and select suitable answer for it :



- (1) It is a morula stage
- (2) The embryo with 8 to 16 blastomeres is called a morula
- (3) Its cells called blastomeres which form by cleavage
- (4) All of these

114. Which changes does not occur during parturition?

- (1) Relaxin is secreted
- (2) Oxytocin is secreted
- (3) Progesterone secretions stop
- (4) Pubic symphysis become tight

115. The signals for parturition originates from :-
 (1) Fully developed foetus
 (2) Placenta
 (3) Both (1) and (2)
 (4) Ovary
116. Select the hormone that is not secreted by placenta: -
 (1) hCG (2) Estrogen
 (3) Progesterone (4) LH
117. Zona pellucida is the :-
 (1) Primary egg membrane
 (2) Secondary egg membrane
 (3) Tertiary egg membrane
 (4) Cellular membrane
118. Synthesis of second polar body :-
 (1) In ovary after completion of meiosis-I
 (2) In oviduct after completion of meiosis-I
 (3) In ovary after completion of meiosis-II
 (4) In oviduct after completion of meiosis-II
119. The glandular tissue of each breast is divided into:-
 (1) 15-20 mammary lobes
 (2) 15-20 alveoli
 (3) 15-20 tubules
 (4) All of the above
120. In which of the following contraceptive method progesterone is not used?
 (1) i-Pill (2) Mala-D
 (3) L.N.G-20 (4) Barrier method
121. Find out the correct sequence of spermatogenesis?
 I. Spermatid
 II. Spermatogonia
 III. Secondary spermatocyte
 IV. Primary spermatocyte
 V. Sperms
 (1) II→I→V→III→IV (2) II→IV→III→I→V
 (3) II→III→IV→I→V (4) II→IV→III→V→I
122. Lactating mother generally does not conceive, because of :-
 (1) Synthesis of prostaglandins
 (2) Presence of large amount of IgA in milk
 (3) High level of oxytocin
 (4) Low level of gonadotropins
123. The Glandular layer of uterus is :-
 (1) Perimetrium (2) Myometrium
 (3) Endometrium (4) Fibrometrium
124. The layer of uterus that disintegrates and regenerates again and again is :-
 (1) Tunica properia (2) Myometrium
 (3) Endometrium (4) Perimetrium
125. Which pair is not incorrect ?
 (1) A.B.P – Secreted by Leydig's cells
 (2) L.H – Acts on sertoli cells
 (3) F.S.H – Secreted by sertoli cells
 (4) Inhibin – Secreted by subtentacular cells and mainly acts of FSH
126. How many secondary spermatocytes are required to form 400 million spermatozoa ?
 (1) 50 million (2) 100 million
 (3) 200 million (4) 400 million
127. How many of the following structures are present only in female?
 Vas deferens, Anus, Bartholin gland, Clitoris, Urethra, Ureter
 (1) 5 (2) 4 (3) 3 (4) 2
128. A female of 38 years is suffering from endometrial cancer, doctor advise her hysterectomy. If she goes through surgery than which of the following changes occurs in her menstrual cycle?
 (1) Ovarian changes (2) Uterine changes
 (3) Menstrual cycle (4) Fertilization

129. Hormonal I.U.D is :-

- (1) LNG-20 (2) Multiload-375
(3) Saheli (4) Lippes loop

130. Which of the following statement is incorrect regarding vagina?

- (1) Fibromuscular organ
(2) Glandular
(3) Female copulatory organ
(4) Partially covered by mucus membrane called hymen

131. After oophorectomy the hormonal level in female blood will be :-

- (1) Estrogen ↑, Prog ↑, FSH ↓, LH ↓
(2) Estrogen ↓, Prog ↓, FSH ↑, LH ↑
(3) Estrogen ↑, Progesterone ↑, FSH ↑, LH ↑
(4) Estrogen ↓, Progesterone ↓, FSH ↓, LH ↓

132. A female of 43 years have 3 identical children, Puberty attains at the age of 11 year and duration of M.C is 30 days. Find the total number of menstrual cycles?

- (1) 384 (2) 339 (3) 369 (4) 381

133. Blastopore is the pore of :-

- (1) Blastocoel (2) Archenteron
(3) Coelom (4) Alimentary canal

134. Which hormone is responsible for maintenance of corpus luteum during pregnancy?

- (1) Progesterone (2) Estrogen
(3) LH (4) H.C.G

135. Which kind of uterine changes occurs during pre-ovulatory phase of menstrual cycle?

- (1) Endometrial hyperplasia
(2) Secretory changes
(3) Development of endometrial glands
(4) Theca interna degenerates

SECTION-B

This section will have 15 questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

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136. Which hormone analogue is injected in the female cattle for super ovulation ?

- (1) LH (2) FSH (3) GH (4) TSH

137. Which type of cross should be done between the animals to combine the qualities of two different breeds?

- (1) Inbreeding
(2) Out-crossing
(3) Cross-breeding
(4) Interspecific hybridization

138. Match the terms given in column I with their descriptions given in column II and select option from the codes given below.

Column I		Column II	
A	Out-crossing	(i)	Mating of closely related individuals within the same breed
B	Interspecific hybridisation	(ii)	Mating of animals of same breed but having no common ancestors on either side of their pedigree for 4-6 generations
C	Cross breeding	(iii)	Mating of animals of two different species
D	Inbreeding	(iv)	Mating of animals belonging to different breeds

(1) A-ii, B-iii, C-iv, D-i (2) A-iii, B-ii, C-iv, D-i

(3) A-iv, B-ii, C-iii, D-i (4) A-ii, B-iv, C-iii, D-i

139. Which of the following statements are true about menstruation ?

- (i) In a normal 28 days menstrual cycle you would expect menstruation to last apporoximately 4-5 days.
- (ii) During menstruation, the myometrium is shed
- (iii) During menstruation, the endometrium is shed
- (iv) Absence of menstruation may indicates an active pregenacy

Options :

- (1) (i), (iii), (iv) are true
- (2) (ii), (iii) are true
- (3) (i), (ii) are true
- (4) (i), (iii) are true

140. Match column-I with Column-II and choose the correct option.

	Column-I		Column-II
a.	Menstruation	(i)	Estrogen
b.	Proliferation of uterine endometrium	(ii)	Low progesterone level
c.	Maintenance of uterine endometrium	(iii)	High progesterone level
d.	Maintenanace of corpus luteum before implantation of blastocyst	(iv)	Luteinising hormone

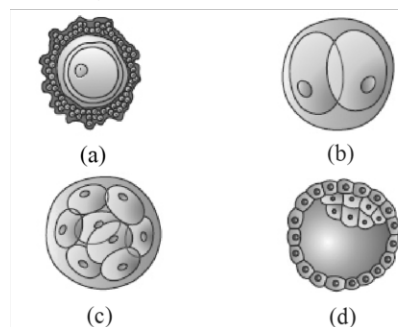
- (1) a(ii), b(i), c(iii), d(iv)
- (2) a(i), b(ii), c(iii), d(iv)
- (3) a(iii), b(ii), c(i), d(iv)
- (4) a(iv), b(i), c(ii), d(iii)

141. These are some events associated with reproduction and embryonic development. Arrange these from first to last and select the correct answer.

- (a) Start formation of corpus luteum
- (b) Cleavage
- (c) Fertilisation
- (d) Ovulation
- (e) Implantation
- (f) Formation of Graafian follicle
- (g) Blastocyst
- (h) Morula
- (i) Zygote

- (1) f, d, a, c, i, b, h, g, e
- (2) f, d, i, c, a, b, h, g, e
- (3) a, f, d, c, b, i, h, g, e
- (4) i, h, g, f, d, c, b, e, a

142. The figure below show four stage (a, b, c, d) of human development. Select the option giving correct identification together with site of occurrence ?



		Developmental stage	Site of occurrence
(1)	(a)	Fertilised egg	Isthmus part of fallopian tube
(2)	(b)	Zygote	End part of fallopian tube
(3)	(c)	Morula	Starting point of Fallopian tube
(4)	(d)	Blastocyst	Uterine cavity

143. Which statement is incorrect :

- (1) Leydig cell secrete testicular hormones called androgen
- (2) The male sex accessory duct include rete testis, vasa efferentia.
- (3) The epididymis leads to vas deferens that ascends to the abdomen and loops over the kidney
- (4) Each testicular lobule contains 1 to 3 highly coiled seminiferous tubule

144. Given below is a list of Hormones may be present in female body :

hCG, progesterone, Estrogen, Thyroxine, Relaxin, hPL, Prolactin

How many of these hormone will present in blood of a pregnant female :-

- (1) 3 (2) 5 (3) 4 (4) 7

145. Which extraembryonic membrane in humans prevents desiccation of the embryo inside the uterus?

- (1) Yolk sac (2) Amnion
- (3) Chorion (4) Allantois

146. Arrange following events of human female reproductive cycle in sequence:-

- (I) Secretion of FSH
- (II) Growth of corpus luteum
- (III) Growth of follicles
- (IV) Ovulation
- (V) Sudden increase in level of L.H.
- (VI) Formation of corpus albicans

- (1) III → I → IV → II → VI → V
- (2) I → III → V → IV → II → VI
- (3) I → IV → III → V → II → VI
- (4) II → I → III → IV → VI → V

147. How many of the following structure are found in fallopian tube?

Morula, secondary oocyte, primary oocyte, Polar body-I, Secondary follicle, Ovum, Zygote, Blastula, gastrula.

- (1) 2 (2) 3 (3) 4 (4) 5

148. Find out the incorrect match for the following table :-

	Column-I	Column-II	Column-III
(i)	Dartos muscle	Voluntry	Layer of scrotum
(ii)	Tunica propia	W.F.C.T	Covering of seminigerous tubule
(iii)	Estrogen	Developing follicle	Maintain thickness of endometrium
(iv)	Germinal epithelium	Cuboidal cell	Layer of ovary and seminiferous tubules

- (1) (i), (ii), (iv) (2) (ii), (iv)
 (3) (ii), (iii) (4) (ii), (iii), (iv)

149. Which of the following statement is incorrect about cleavage?

- (1) Interphase is very short
- (2) Number of blastomere increases
- (3) Size of embryo increases
- (4) Only S-phase present in interphase

150. Which of the following can not be detected in a developing foetus by amnio centesis?

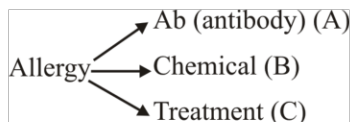
- (1) Sex of foetus
- (2) Down syndrome
- (3) Metabolic disorder
- (4) Cleft palate

Topic : HUMAN HEALTH AND DISEASE (COMPLETE), EVOLUTION (TILL TAUGHT), HUMAN REPRODUCTION, ANIMAL HUSBANDRY**SECTION-A**

TG: @Chalnaayaaar

Attempt All 35 questions

151.



- (1) A–IgA, B–Heparin, C–Steroids
 (2) A–IgA, B–Steroids, C–Histamine
 (3) A–IgE, B–Histamine, C–Steroids
 (4) A–IgG, B–Steroids, C–Histamine

152. Immune deficiency disorder, which occur due to mutation in gene, involved in maturation of T and B lymphocytes is

- (1) IDDM (2) Hashimoto disease
 (3) SCID (4) Pernicious anemia

153. Which is common between AIDS and Hepatitis-B ?

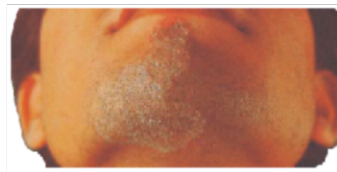
- (i) Both are viral diseases
 (ii) Both are transmitted by infected needles and syringes
 (iii) Both are infectious diseases
 (iv) Both are chronic diseases
 (v) Both are ultimately fatal

- (1) ii, iii, iv and v (2) i, ii, iii and v
 (3) i, ii, iii, iv and v (4) ii, iii and v

154. Infectious stages of plasmodium for female Anopheles mosquitoes is __A__ and for Human is __B__

- (1) B-Gametocytes, A-Sporozoites
 (2) B-Cryptozoites, A-Cryptomerozoites
 (3) B-Sporozoites, A-Gametocytes
 (4) B-Erythrocytes, A-Gametocytes

155. Which statement is correct about given diagram?



- (a) Microsporum, Trichophyton and Epidermophyton are responsible for this.
 (b) Allergic reactions.
 (c) Caused by Wuchereria bancrofti
 (d) Heat and moisture help these to grow

- (1) a, c, d (2) b, c, d
 (3) a, d (4) All of these

156. Which of the following is not an early symptom of sexually transmitted infections (STI) ?

- (1) Fluid discharge
 (2) Itching
 (3) Ectopic pregnancy
 (4) Slight pain

157. Which of the following drug interferes with transport of dopamine & produce euphoria :-

- (1) Benzodiazepine (2) Hashish
 (3) LSD (4) Coca alkaloid

158. The first cellular form of life possibly originated about :

- (1) 2 million years ago
 (2) 20 million years ago
 (3) 200 million years ago
 (4) 2000 million years ago

159. Which theory supported the idea that life originates from pre-existing life ?
- (1) Abiogenesis
 - (2) Biogenesis
 - (3) Special creation theory
 - (4) Modern theory
160. The first organisms which released oxygen in the environment were :-
- (1) Cyanobacteria
 - (2) Sulphur bacteria
 - (3) Iron bacteria
 - (4) Planktonic sulphur bacteria
161. S.L. Miller an american scientist created electric discharge in a closed flask containing __ (A) __ at __ (B) __.
- (1) A-CH₃, NH₃, H₂ and H₂O vapour, B-800°C
 - (2) A-CH₄, H₂, NH₃ and H₂O vapour, B-600°C
 - (3) A-CH₄, H₂, NH₃ and H₂O vapour, B-800°C
 - (4) A-CH₄, CO₂ and H₂O vapour, B-800°C
162. The ideas of theory of special creation were strongly challenged during 19th century by:
- (1) Louis Pasteur (2) A. I. Oparin
 - (3) Stanley Miller (4) Charles Darwin
163. The evolution can be best explained by :
- (1) Theory of inheritance of acquired characters
 - (2) Descent with modification
 - (3) Theory of continuity of germplasm
 - (4) Recapitulation theory
164. Study of history of life forms on earth is called :-
- (1) Exobiology
 - (2) Cosmology
 - (3) Evolutionary biology
 - (4) Ethology
165. Life cannot originate from inorganic materials at present because :
- (1) High degree of environmental pollution
 - (2) A very high amount of oxygen in the atmosphere
 - (3) Absence of raw materials.
 - (4) Very high atmospheric temperature
166. Palaeontological evidences for evolution refer to the :
- (1) Development of embryo
 - (2) Homologous organs
 - (3) Fossils
 - (4) Analogous organs.
167. Which of the following is the most appropriate sequence for types of evolution during early age of earth ?
- (1) Organic → Chemical → Biological
 - (2) Biological → Chemical → Organic
 - (3) Chemical → Biological → Organic
 - (4) Chemical → Organic → Biological
168. Which of the following affects both physical and mental health?
- (1) Balanced diet
 - (2) Regular exercise
 - (3) Yoga
 - (4) Drug and alcohol abuse

169. How Many of these diseases are transmitted by insect vectors ?
Malaria, AIDS, Dengue, Filariasis, Common cold.
- (1) Four (2) Three
(3) Two (4) One
170. Antigen presenting cells have
- (1) MHC–I (2) MHC–II
(3) Both 1 and 2 (4) None
171. Which of the following cells release perforin protein while attacking pathogen ?
A. NK–Cell,
B. B–Plasma cell,
C. T–Killer cell
- (1) A and B (2) A, B and C
(3) A and C (4) B and C
172. Immune response on 3rd encounter of same antigen is-
- (1) Amnestic Immune response
(2) Secondary Immune response
(3) Tertiary Immune response
(4) Primary Immune response
173. How many of the following will decrease in case of healthy societies ?
Infant mortality, economic prosperity, longevity, maternal mortality, productivity.
- (1) Four (2) Three
(3) Two (4) One
174. Fever is not a symptom in case of–
- (1) Hay fever (2) Malaria
(3) Typhoid (4) Ascariasis
175. Housefly act as mechanical carrier for which pathogen?
- (1) Amobeiasis
(2) Ascaris
(3) Entamoeba histolytica
(4) Ascariasis
176. Which cancer can be prevented if a girl is given vaccine for human papillomavirus?
- (1) Genital warts (2) Cervical cancer
(3) Breast cancer (4) Lung cancer
177. Which of the following is a cause of cancer, used for detection of cancer and also used for treatment of cancer ?
- (1) X–rays
(2) Monoclonal antibodies
(3) MRI
(4) Alpha interferon
178. LSD (Lysergic acid diethyl amides) is a dangerous Hallucinogen, obtained from–
- (1) Atropa belladonna (2) Dhatura
(3) Cannabis sativa (4) Claviceps purpurea
179. **Assertion** : Adolescence is a very vulnerable phase of mental and psychological development of an individual.
Reason : Adolescence is bridge linking childhood and adulthood which is accompanied by several biological and behavioural changes.
- (1) Both assertion and reason are true and reason is correct explanation of assertion.
(2) Both assertion and reason are true but reason is not correct explanation of assertion.
(3) Assertion is true, reason is false.
(4) Both assertion and reason are false.

180. A patient undergoing CABG will not require –
- (1) Blood group matching
 - (2) MHC-matching
 - (3) Lifetime immunosuppressive drugs
 - (4) All of the above
181. Which of the following is an autoimmune disorder which affects many people in our society?
- (1) Diabetes mellitus
 - (2) Myasthenia Gravis
 - (3) Rheumatoid arthritis
 - (4) Multiple sclerosis
182. Which of the following cells can suppress both cellular and humoral immunity from attacking own body?
- (1) Immunosuppressive drugs
 - (2) T-suppressor cells
 - (3) T-helper cells
 - (4) Naive B-cells
183. **Statement 1 :** Stellar distances are measured in light years.
Statement 2 : When we see stars we apparently are peeping into the future.
- (1) Both Statement 1 and 2 are true
 - (2) Statement 1 is true, Statement 2 is false
 - (3) Statement 1 is false, Statement 2 is true
 - (4) Both Statement 1 and 2 are false

184. **Assertion :** Universe expanded.
Reason: Temperature of universe came down.
- (1) Both assertion and reason are true and reason is correct explanation of assertion.
 - (2) Both assertion and reason are true but reason is not correct explanation of assertion.
 - (3) Assertion is true, reason is false.
 - (4) Both assertion and reason are false.
185. How old is earth according to theory based on religious literature ?
- (1) 4B years
 - (2) 4.5B years
 - (3) 10,000 years
 - (4) 4000 years

SECTION-B

This section will have 15 questions. Candidate can choose to attempt any 10 question out of these 15 questions. In case if candidate attempts more than 10 questions, first 10 attempted questions will be considered for marking.

186. Which of the following statement is not correct about typhoid ?
- (a) Sustained high fever (39° to 40°F)
 - (b) Weakness
 - (c) Constipation
 - (d) Headache and loss of appetite
 - (e) Coughing
- (1) a, c, d
 - (2) c, d, e
 - (3) a and e only
 - (4) All are correct
187. Which disease is not caused by bacteria ?
- (1) Shigelliasis
 - (2) Plague
 - (3) Diphtheria
 - (4) Elephantiasis

188. Several genes called _____ have been identified in normal cells which when activated will turn into _____ and under certain conditions could lead to cancerous transformation of cells.

Complete the above paragraph by selecting correct sequence of words.

- (1) oncogenes, protooncogenes
- (2) cellular oncogenes, protooncogenes
- (3) protooncogenes, oncogenes
- (4) oncogen, cellular oncogenes

189. How many of the following types of Cancer are associated with smoking ?

Lung Cancer, Urinary bladder, Throat, Prostate, Skin Cancer, Liver Cancer

- (1) Five
- (2) Two
- (3) Three
- (4) Four

190. The scientist related with the theory of biogenesis and experiments with swan-necked flasks is-

- (1) Von Helmont
- (2) Louis Pasteur
- (3) Miller
- (4) Haeckel

191. Read the given statement carefully and choose the correct option :-

Statement I :- Adaptive radiation is based on divergent evolution.

Statement II :- Homologous organs indicates divergent evolution.

- (1) Statement I correct But II incorrect.
- (2) Statement I and II, Both are incorrect.
- (3) Statement II correct But I incorrect.
- (4) Statement I and II, Both are correct.

192. Following are the examples of some homologous and analogous organs

- a. Tendrils of Cucurbita and thorns of Bougainvillea
- b. Forelimbs of mammals.
- c. Flipper of Penguin and Dolphin.
- d. Wings of insects and wings of birds.

Choose the correct option.

- (1) Homologous organs → a and c
Analogous organs → b and d
- (2) Homologous organs → a and b
Analogous organs → c and d
- (3) Homologous organs → a and d
Analogous organs → b and c
- (4) Homologous organs → c and d
Analogous organs → a and b

193. When more than one adaptive radiations appear in an isolated geographical area, then it is called :

- (1) Natural selection
- (2) Convergent evolution
- (3) Divergent evolution
- (4) Retrogressive evolution

194. Which of the following is part of Cellular Barrier ?

- (1) B-Lymphocytes
- (2) T-Killer lymphocytes
- (3) Large Granular Lymphocytes
- (4) T-Helper Lymphocytes

195. How many intra-chain disulphide bonds are present in H₂L₂ ?

- (1) 4
- (2) 16
- (3) 8
- (4) 12

196. **Assertion :** Mind can influence our health.

Reason : Nervous and endocrine systems can influence our immune system.

- (1) Both assertion and reason are true and reason is correct explanation of assertion.
- (2) Both assertion and reason are true but reason is not correct explanation of assertion.
- (3) Assertion is correct, reason is false.
- (4) Both assertion and reason are false.

197. Correct sequence in which enzymes of HIV work during HIV replication is-

- (1) Protease – reverse transcriptase – integrase
- (2) Reverse transcriptase – protease – integrase
- (3) Integrase – reverse transcriptase – protease
- (4) Reverse transcriptase – integrase – protease

198. In a standard ECG, occurrence of joint diastole will be best represented by which of the following option?

- (1) Between Q and S (2) Between P and QRS
- (3) Between T and P (4) Between P and T

199. Evolution does not mean hidden potentials.

- (1) Unrevealing (2) Unfolding
- (3) Unrolling (4) Both 2 and 3

200. Which of the following is still a favorite idea for some astronomers regarding origin of life ?

- (1) Oparin-Haldane theory
- (2) Autogenesis
- (3) Panspermia
- (4) Biogenesis

Space for Rough Work

Read carefully the following instructions :

1. Each candidate must show on demand his/her Allen ID Card to the Invigilator.
2. No candidate, without special permission of the Invigilator, would leave his/her seat.
3. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty.
4. Use of Electronic/Manual Calculator is prohibited.
5. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
6. No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
7. The candidates will write the Correct Name and Form No. in the Test Booklet/Answer Sheet.

ALLEN CAREER INSTITUTE Pvt. Ltd.

Registered & Corporate Office : 'SANKALP', CP-6, Indra Vihar, Kota (Rajasthan) INDIA-324005

Ph. : +91-744-3556677, +91-744-2757575 | E-mail : info@allen.ac.in | Website : www.allen.ac.in